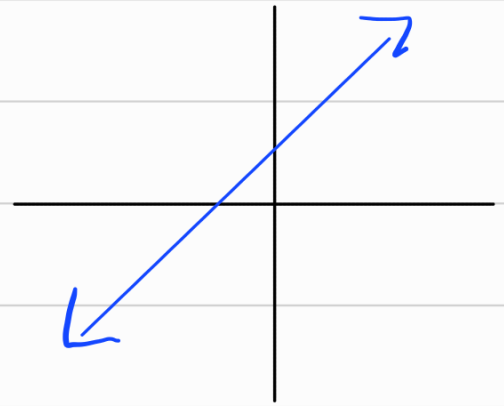


Linear functions

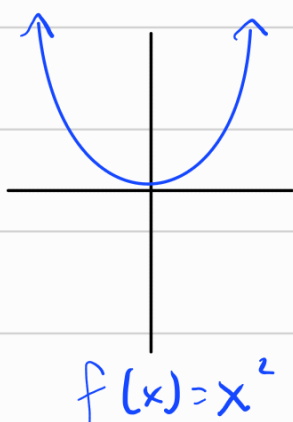
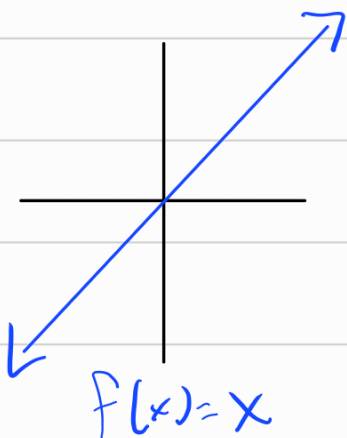
$$y = f(x) = \underbrace{m}_\text{slope} x + \underbrace{b}_\text{y-intercept}$$



Domain	$\mathbb{R}$
Range	$\mathbb{R}$
Type	odd if $b=0$
Inc/dec	inc if $m > 0$, dec if $m < 0$
end behavior	$\lim_{x \rightarrow \infty} = \pm \infty$, $\lim_{x \rightarrow -\infty} = \mp \infty$ depends on m

Power functions

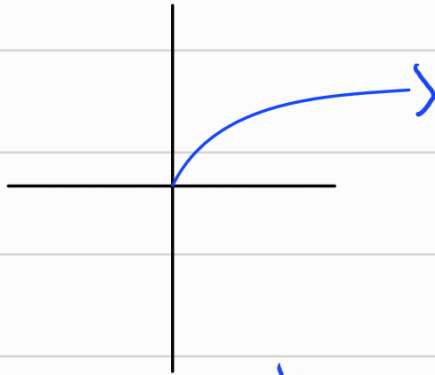
$f(x) = x^n$ for $n > 0$ and n an integer



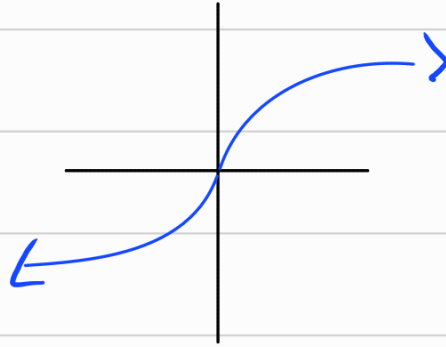
	n even	n odd
Domain	$\mathbb{R}$	$\mathbb{R}$
Range	$[0, \infty)$	$\mathbb{R}$
Type	even	odd
Inc/dec	inc $[0, \infty)$ dec $(-\infty, 0]$	inc
end behavior	$\lim_{x \rightarrow \infty} f(x) = \infty$ $\lim_{x \rightarrow -\infty} f(x) = \infty$	$\lim_{x \rightarrow \infty} f(x) = \infty$ $\lim_{x \rightarrow -\infty} f(x) = -\infty$

Root functions

$$f(x) = x^{1/n} = \sqrt[n]{x}$$



$$f(x) = x^{1/2}$$



$$f(x) = x^{1/3}$$

	n even	n odd
Domain	$[0, \infty)$	$\mathbb{R}$
Range	$[0, \infty)$	$\mathbb{R}$
Type	—	odd
Inc/dec	inc $[0, \infty)$	inc
end behavior	$\lim_{x \rightarrow \infty} f(x) = \infty$	$\lim_{x \rightarrow \infty} f(x) = \infty$ $\lim_{x \rightarrow -\infty} f(x) = -\infty$

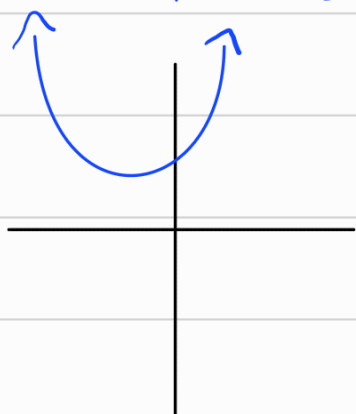
Polynomials

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

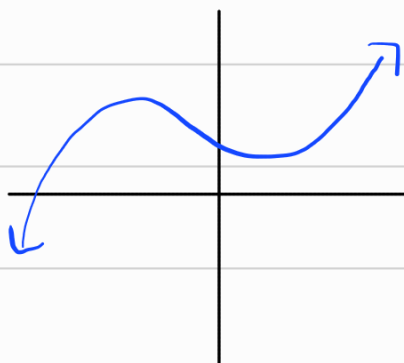
$a_i \in \mathbb{R}$ are real numbers

the "degree" of a polynomial is largest power of x

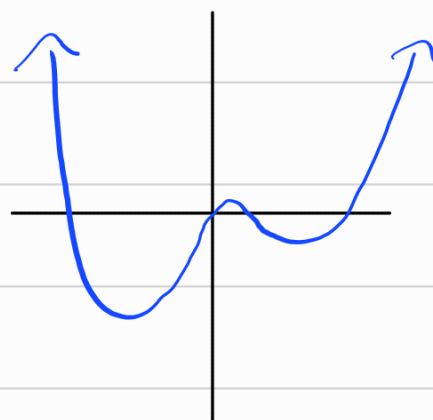
$f(x) = 3x^3 - 2x^2 + x - 1$ has degree 3



$$f(x) = x^2 + x + 1$$



$$f(x) = x^3 - x + 1$$



$$f(x) = x^4 - 3x^2 + x$$

	degree even	degree odd
Domain	$\mathbb{R}$	$\mathbb{R}$
Range	Depends	$\mathbb{R}$
Type	In general, neither $\rightarrow$	
Inc/dec	Changes on different intervals $\rightarrow$	
end behavior	$\lim_{x \rightarrow \infty} f(x) = \pm \infty$ $\lim_{x \rightarrow -\infty} f(x) = \pm \infty$ (Same on both sides)	$\lim_{x \rightarrow \infty} f(x) = \pm \infty$ $\lim_{x \rightarrow -\infty} f(x) = \mp \infty$ (Flipped on other side)

Rational functions

Say $P(x)$ and $Q(x)$ are polynomials,

A rational function is of the form

$$f(x) = \frac{P(x)}{Q(x)}$$

We can't say a lot about these functions in general.

However, domain is interesting as we can't divide by 0.

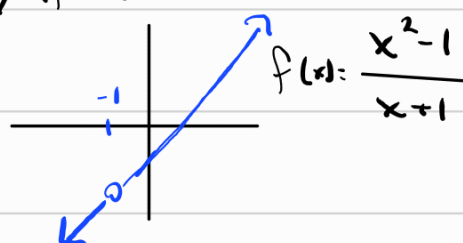
Say $Q(x)$ has roots $\alpha_1, \dots, \alpha_n$ (i.e. $Q(\alpha_i) = 0$)

Then, $\text{Domain}(f(x)) = \mathbb{R} \setminus \{\alpha_1, \dots, \alpha_n\} = \{x \in \mathbb{R} \mid Q(x) \neq 0\}$

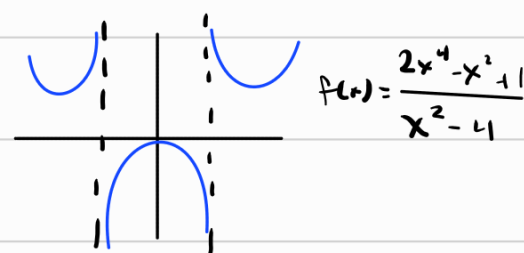
↳ all real numbers s.t. $Q(x) \neq 0$

These functions often have holes or asymptotes

Hole: when $Q(\alpha) = 0$ and $P(\alpha) = 0$ as well



Asymptote: Caused when $Q(\alpha) = 0$ but $P(\alpha) \neq 0$



Algebraic functions

A function is called Algebraic if it can be constructed via algebraic operations (add, sub, multiply, divide, roots, etc.)

Ex: $f(x) = \sqrt{x-2}$, $f(x) = \frac{3x^4 - 2x^2 - 4}{\sqrt{x^2 - 11} + 8}$

These functions are often very complicated, but we can often easily find their domain by keeping two things in mind

- 1) Don't divide by 0
- 2) Don't even root negatives

Ex: $f(x) = \frac{x^4 - 16x^2}{3 - \sqrt{x}}$

We need $x \geq 0$ since $\sqrt{x}$ has domain $[0, \infty)$

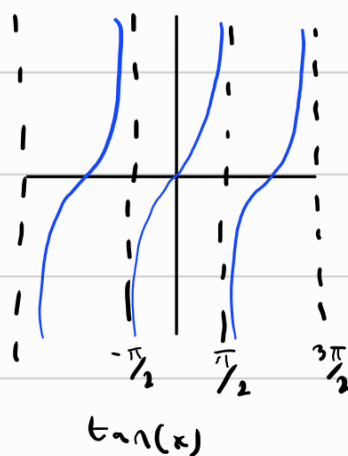
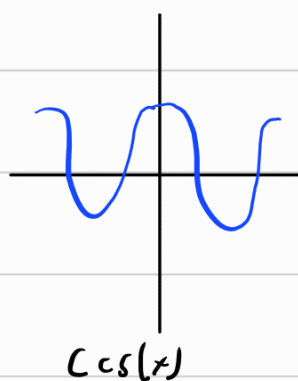
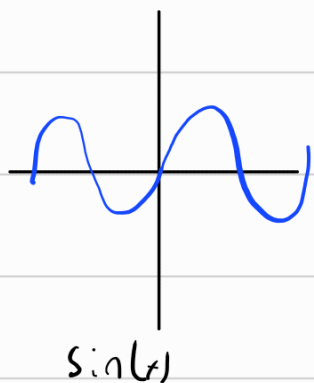
We cannot have $x=9$ as $3 - \sqrt{x} = 0$ if $x=9$

So

$$\text{Domain}(f(x)) = [0, 9) \cup (9, \infty)$$

Trigonometric Functions

The big three are $\sin(x)$, $\cos(x)$, $\frac{\sin(x)}{\cos(x)}$
 $\tan(x)$



	$\sin(x)$	$\cos(x)$	$\tan(x)$
Domain	$\mathbb{R}$	$\mathbb{R}$	$\{x \in \mathbb{R} \mid x \neq \frac{\pi}{2} + k\pi\}$
Range	$[-1, 1]$	$[-1, 1]$	$\mathbb{R}$
Type	odd	even	odd
Inc/dec	Changes	Changes	Increasing
end behavior	Undefined		

Ex: Find domain of $f(x) = \frac{1}{1-2\cos(x)}$

We just need to exclude when $1-2\cos(x)=0$

i.e. $\cos(x) = \frac{1}{2}$, so all x in $\mathbb{R}$ except $x = \frac{\pi}{3} + 2k\pi, \frac{5\pi}{3} + 2k\pi$

Exponential functions

$$f(x) = b^x \quad \text{for } b > 0$$

The function behaves different based on the value of b .

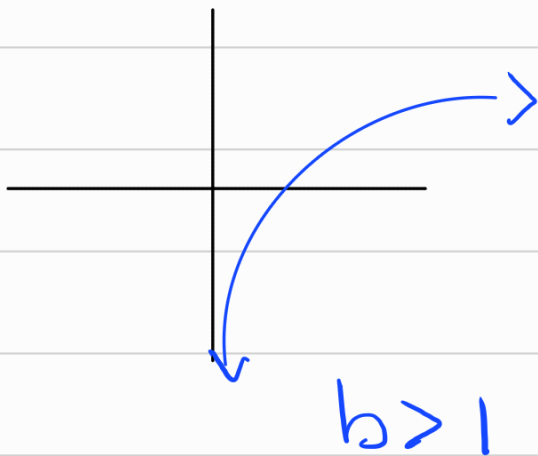


	$b > 1$	$0 < b < 1$
Domain	$\mathbb{R}$	$\mathbb{R}$
Range	$[0, \infty)$	$[0, \infty)$
Type	_____	_____
Inc/dec	Inc	Dec
end behavior	$\lim_{x \rightarrow \infty} f(x) = \infty$ $\lim_{x \rightarrow -\infty} f(x) = 0$	$\lim_{x \rightarrow \infty} f(x) = 0$ $\lim_{x \rightarrow -\infty} f(x) = \infty$

Logarithmic functions

$$f(x) = \log_b(x) \quad b > 0$$

Inverse to exponential function. So $\log_b(b^x) = x$



	$b > 1$	$0 < b < 1$
Domain	$[0, \infty)$	$[0, \infty)$
Range	$\mathbb{R}$	$\mathbb{R}$
Type	_____	_____
Inc/dec	Inc	Dec
end behavior	$\lim_{x \rightarrow \infty} f(x) = \infty$ $\lim_{x \rightarrow 0} f(x) = -\infty$	$\lim_{x \rightarrow \infty} f(x) = -\infty$ $\lim_{x \rightarrow 0} f(x) = \infty$

} reverse of
exponential
functions